

Particulate Matter Monitoring & Health Effects

Issue 2026-09-16 | PubMed entry window 16 September 2026 | 21 records screened in scope

21RECORDS IN SCOPE
(27 EXCLUDED, 3 HELD)**9**OF 10 SUBTOPIC
BANDS POPULATED**6**EFFECT ESTIMATES
(ALL WITH A CI)**5**TIER-A
RECORDS**0**CONSENSUS RECORDS
ADMITTED

Signal of the day

Landscape-fire $PM_{2.5}$ is 19% of ambient mass and 31% of the attributable deaths — and conventional epidemiology has been understating it by more than half.

Ju et al. (*Lancet Planet Health*) linked daily mortality in **1,366 communities across ten countries and territories**, 2000–19, **47,844,851 deaths**, to population-weighted landscape-fire $PM_{2.5}$, and estimated the association with instrumental-variable models rather than the usual time-series regression. Each $1 \mu\text{g}/\text{m}^3$ of 2-day mean fire $PM_{2.5}$ raised mortality **0.60% (95% CI 0.19–1.00)** and mortality-related economic losses **0.59% (0.21–0.97)**, giving **906,869 attributable deaths** and **US\$994.5 billion**. Two numbers carry the issue. First, the disproportion: fire-sourced particles are **19.0% of ambient $PM_{2.5}$ mass but 30.6% of $PM_{2.5}$ -attributable deaths** — roughly a **1.9-fold per-unit potency premium** over background mass. Second, the method gap: the IV estimates exceeded conventional association estimates by a **median 56.0% (2.2–111.3)** for mortality and **62.3% (9.0–116.1)** for economic loss, i.e. the standard design is biased *toward the null* here, not away from it. The instrument itself is not named in the abstract, and exclusion-restriction validity against co-emitted gases and weather carries the whole causal claim, so this is a strong result resting on one unexamined assumption. For instrumentation the implication is blunt: a network reporting total $PM_{2.5}$ mass cannot see the fraction that carries a third of the deaths.

What changed today

- ▶ **Two independent causal-inference papers both found the tail, not the mean, carries the burden.** Ju et al. show fire-sourced PM_{2.5} is potency-enriched relative to its mass share. Bao et al. (*J Environ Manage*), evaluating 88 Chinese clean-heating pilot cities with an augmented synthetic control, found the policy moved the heating-season *mean* only $-3.01 \mu\text{g}/\text{m}^3$ but P95 by -5.76 , P99 by -5.96 and exceedance days by -7.92 ; high-pollution days were 27.8% of improvement days and 48.8% of the mortality difference. A mean-only evaluation of either paper's exposure would have halved its own answer. Both argue for networks resolved enough to capture episode tails.
- ▶ **Aerosol-layer height and fluorescence lidar both landed in the same window, and the second undermines a standing sensor assumption.** Li et al. (*AMT*) retrieve aerosol layer height from HARP2/PACE multi-angle polarimetry at **RMSE 1.03 km** against EarthCARE ATLID — the vertical term that dominates AOD-to-surface-PM error for smoke and dust. Hu et al. (*ACP*), observing 2023 Canadian wildfire smoke at two sites with different fluorescence lidar designs, report **near-absence of hygroscopic growth at 90–100% RH**. If that generalises, every κ -Köhler humidity correction applied to optical low-cost PM sensors is systematically wrong during exactly the episodes those sensors are deployed to catch.
- ▶ **A clean PM_{2.5} null, which the archive needs more of.** Ammons et al. (*Environ Res*) meta-analysed seven studies of PM_{2.5} and gynaecologic cancer: ovarian **HR 1.02 (0.91–1.13)** per $5 \mu\text{g}/\text{m}^3$, and **0.94 (0.81–1.09)** restricted to co-pollutant-adjusted studies. The endometrial estimate is nominally positive but its confidence interval $-1.01 (1.00–1.01)$ — is implausibly tight for five studies and no I^2 is reported, which points to shared cohorts rather than precision.
- ▶ **Consensus contributed nothing admissible for the first time.** The sweep returned 10 calibration records and 10 fusion records; **seven were already in the archive by title, and every one of the five that survived title screening failed a Crossref created check** — 2026-01-02, 2026-05-08, 2026-05-27, 2026-08-22 and 2025-12-23. See the provenance note.
- ▶ **Europe PMC's AGRICOLA leg was found to retro-index.** Two records returned inside a CREATION_DATE window of 16 September resolve to **pubYear 2021 and 2014**. This is a source defect, not a screening miss, and it is the reason the AGR source now requires a publication-year check before any record from it is summarised.
- ▶ **Indoor and appliance-side exposure got two usable methods papers.** Tam et al. (*Curr Environ Health Rep*) find nonzero emissions for **at least 41 toxic air contaminants** from gas appliances, with formaldehyde and benzene emission factors spanning **two orders of magnitude** — and suggest CH₄ and CO as inference surrogates, which is a directly implementable low-cost sensing target. Yoshizaki et al. (*J Am Coll Surg*), in a 64-patient randomised trial, cut particle counts **96–97%** with an 8 mm capture distance and found no significant effect at 18 mm — a 10 mm change in near-field geometry deciding the whole result.

Corpus analytics

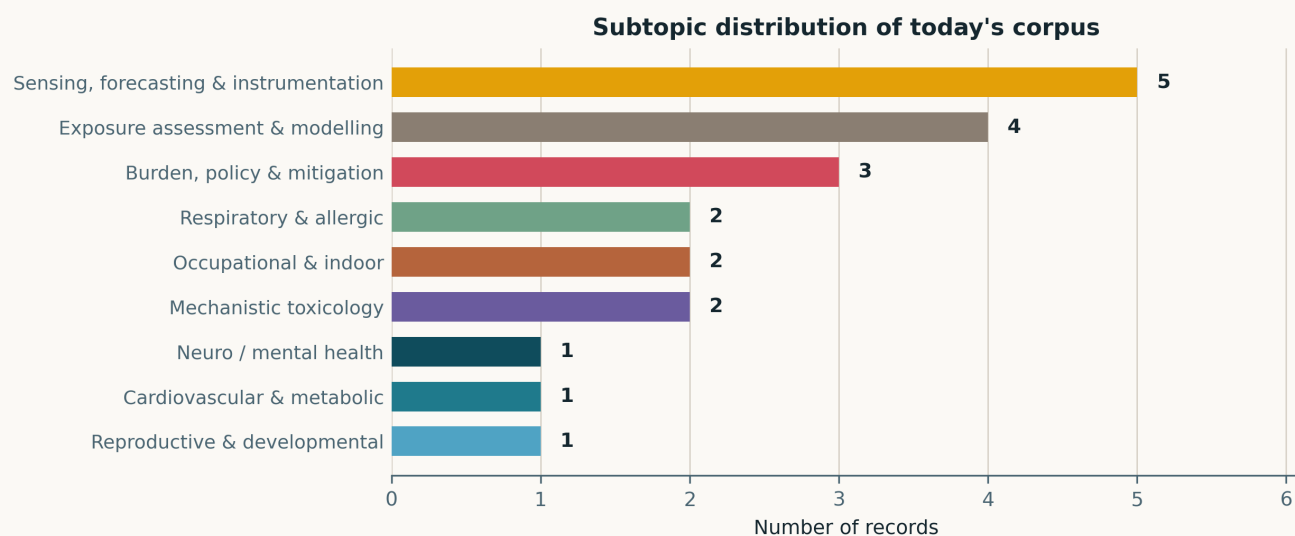


Fig. Subtopic assignment of the 21 in-scope records; single-label by dominant contribution. *Sensing, forecasting & instrumentation* leads with 5 and *Exposure assessment & modelling* follows with 4, so the two measurement-side bands together hold 9 of 21 — the instrumentation axis outweighs any single health axis today. *Burden, policy & mitigation* holds 3, three bands hold 2 (respiratory, occupational/indoor, mechanistic) and three hold a single record each. *Other clinical endpoints* is the one empty band.

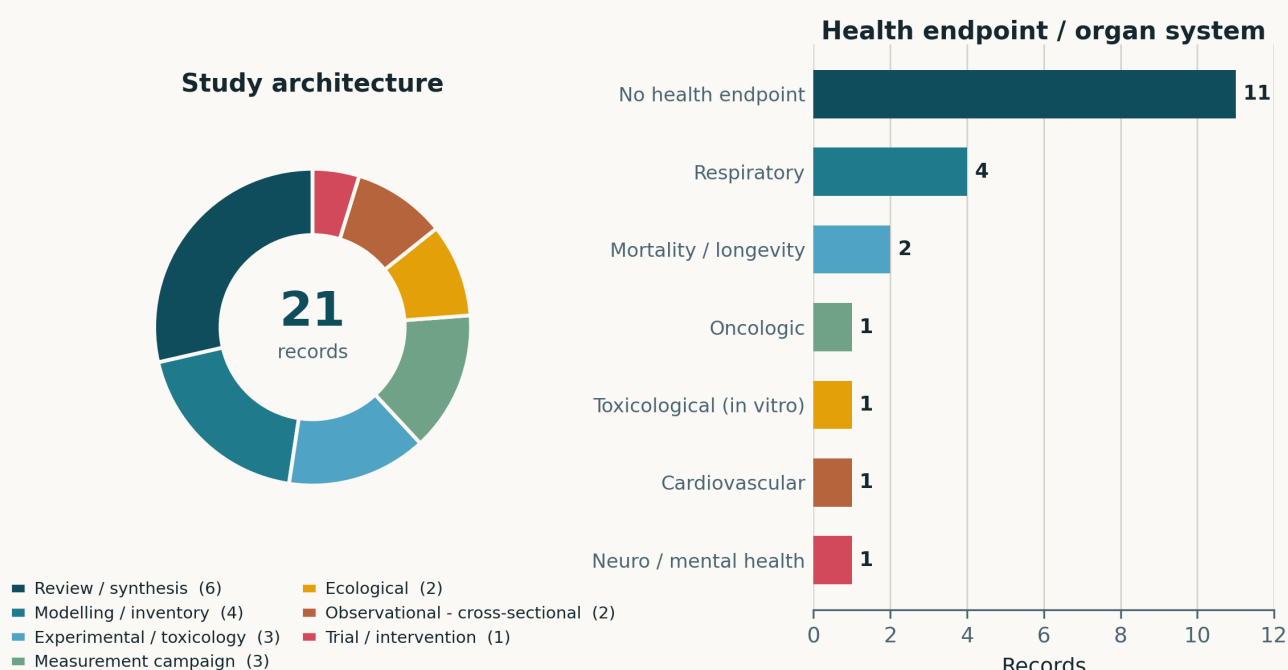


Fig. Left — study architecture. *Review / synthesis* is the largest group at 6 of 21 (29%), an unusually review-heavy day; *Modelling / inventory* holds 4, *Experimental / toxicology* and *Measurement campaign* 3 each, *Ecological* and *Observational — cross-sectional* 2 each, and there is a single randomised trial. Right — health endpoint. **11 of 21 records carry no human health endpoint at all**, the single largest category and a direct consequence of the measurement-side weighting above; *Respiratory* is the largest health endpoint at 4, and mortality, cardiovascular, neuro/mental-health, oncologic and in-vitro toxicology hold one or two each.

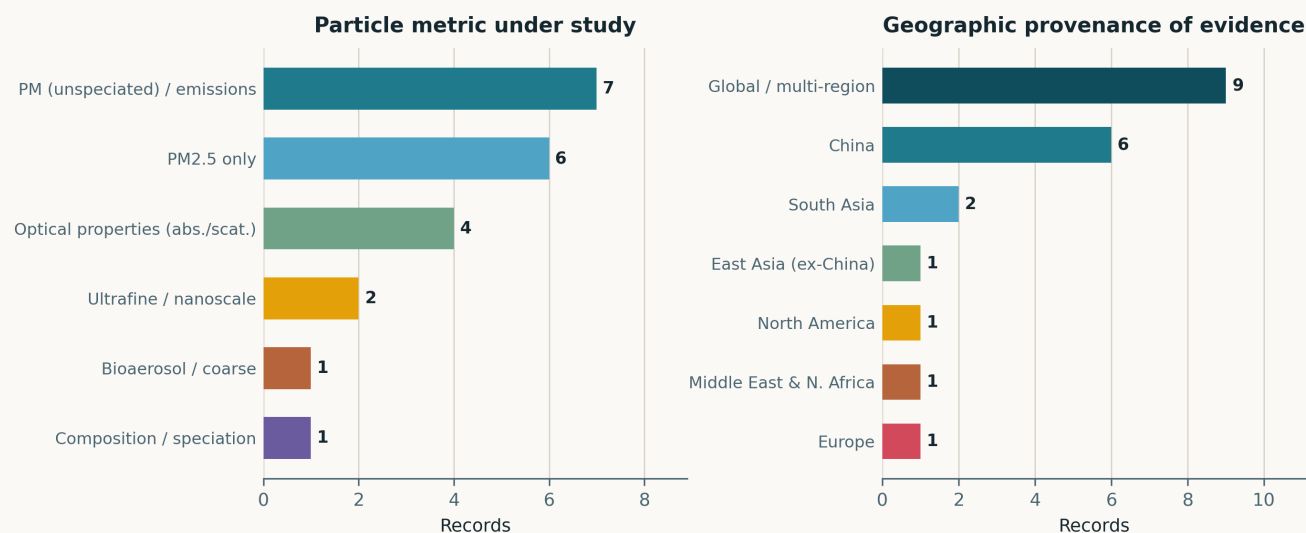


Fig. Left — particle metric. Unspeciated PM and emissions lead at 7, PM_{2.5} alone at 6, and **optical properties (absorption/scattering)** at 4 — the AMT and ACP retrieval records, an unusually large optical share for a daily issue. Ultrafine particles are the explicit subject of 2, bioaerosol and composition 1 each. Right — geographic provenance. Global/multi-region leads with 9, China follows with 6, South Asia holds 2, and Europe, North America, the Middle East & N. Africa and East Asia (ex-China) hold one each. Sub-Saharan Africa and Latin America are absent.

Life-course windows addressed today (records may span several stages)

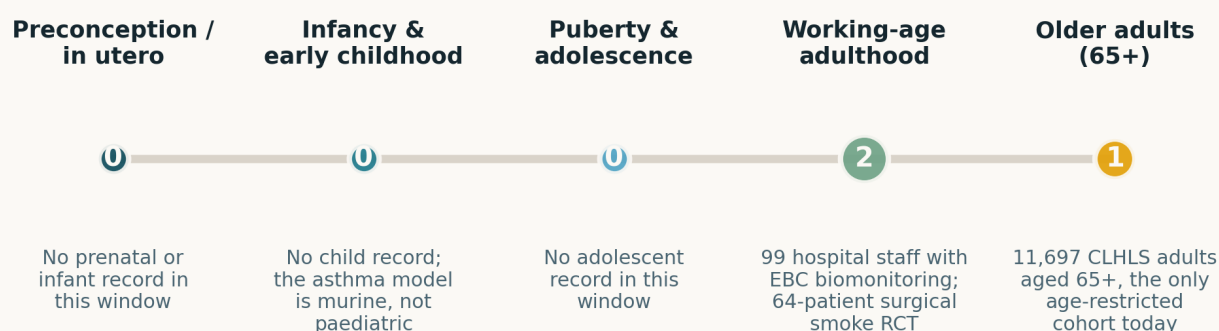


Fig. Life-course windows addressed; counts are non-exclusive. **Three of the five windows are empty today** — no prenatal, childhood or adolescent record appears at all, and the one asthma model in the corpus is murine rather than paediatric. The occupational window holds 2 (hospital-staff breath biomonitoring and the surgical-smoke trial) and older adults 1 (the CLHLS depression cohort). A developmental-exposure gap this complete is unusual and is worth watching across the week rather than reading from one day.

Where the work is happening: subtopic x study architecture

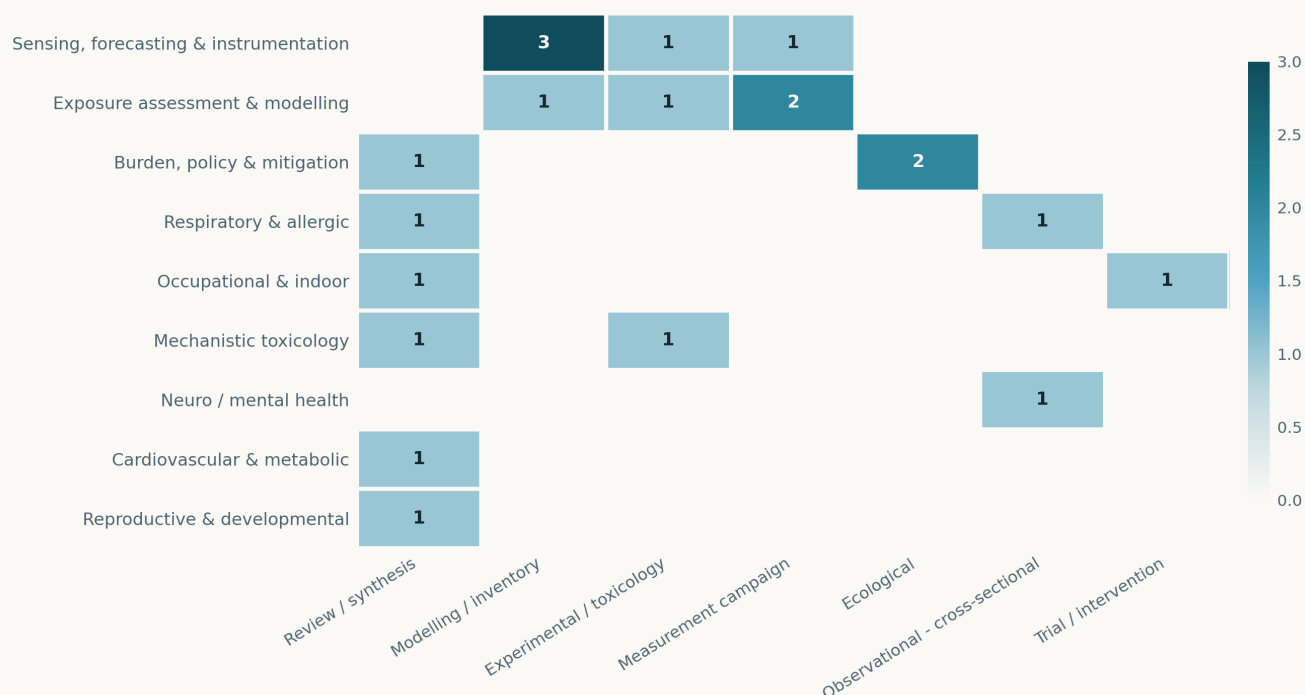


Fig. Subtopic × study-architecture cross-tabulation. Two features are worth naming precisely. First, the measurement–health split is nearly, but not quite, total: all 5 *Sensing* records carry no human health endpoint, and of the 4 *Exposure assessment* records **exactly one** does — the Haifa breath-biomonitoring study, whose endpoint is respiratory. Second, *Mechanistic toxicology* is **not** wholly experimental this issue: one of its two records is a systematic review, so the usual all-experimental pattern is broken. The review-heavy column otherwise sits entirely on the health-side bands.

Quantitative associations reported in today's corpus

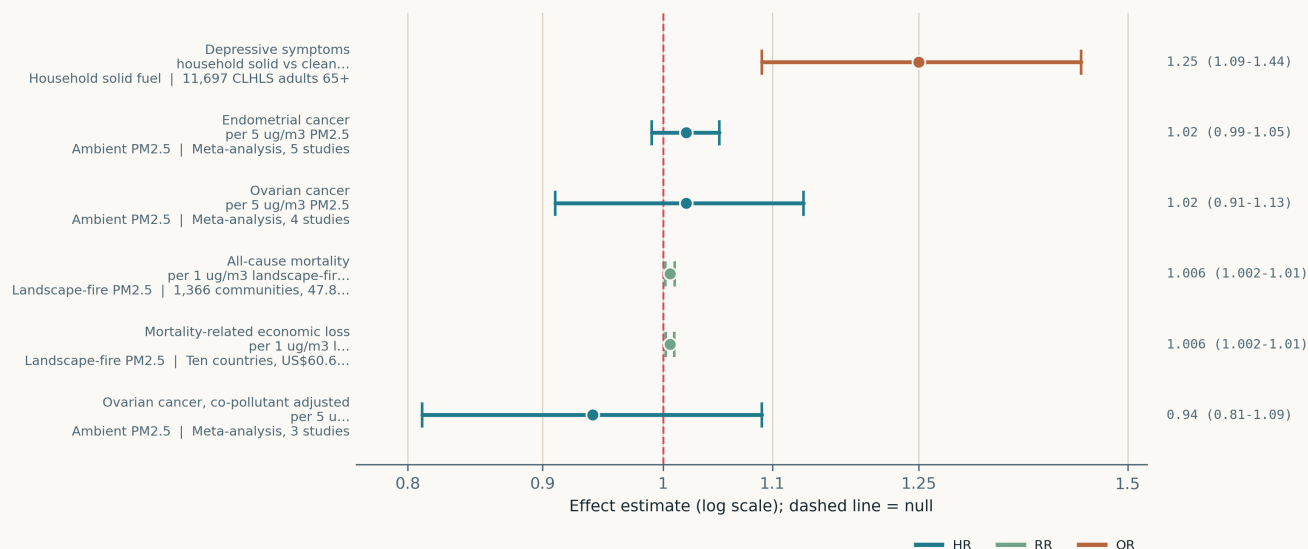


Fig. The 6 interval-bearing estimates in this issue, on a common logarithmic axis. **Exposure contrasts are not mutually comparable:** the two landscape-fire estimates are per 1 $\mu\text{g}/\text{m}^3$ of a 2-day mean, converted from the published percent changes to risk-ratio equivalents and flagged as such; the gynaecologic-cancer HRs are per 5 $\mu\text{g}/\text{m}^3$ of long-term exposure; and the solid-fuel OR is a binary exposed/unexposed contrast with no measured concentration behind it. Read direction and precision, not relative magnitude. **Three of the six cross unity**, and all three are the gynaecologic-cancer estimates: ovarian 1.02 (0.91–1.13), co-pollutant-adjusted ovarian 0.94 (0.81–1.09) — the only point estimate below 1 in the issue — and endometrial 1.02 (0.99–1.05). The three that exclude unity are the two landscape-fire estimates, whose intervals are extremely tight because they rest on 47.8 million deaths, and the household solid-fuel OR. The endometrial interval is so narrow that it should be read as an artefact of shared cohorts rather than as precision.

Paper-level digest

Ordered by cluster, largest first. Each entry gives the methodological core, the finding that matters and the limitation that actually threatens the inference — not an abstract paraphrase.

Sensing, forecasting & instrumentation

5 records

Li et al. 2026 Atmos Meas Tech

A multi-angle and polarization-based retrieval algorithm for aerosol layer height of smoke and dust

Optimal-estimation retrieval adding multi-angle polarimetry to conventional spectral aerosol-layer-height methods, exploiting polarisation sensitivity to the Rayleigh-versus-aerosol scattering contrast. Information-content analysis quantifies the gain as increased degrees of freedom for signal over radiance-only observation — measured, not asserted. Applied to HARP2 on PACE and validated against ATLID on EarthCARE, **RMSE 1.03 km** across all collocated samples. That is the number to hold: about a kilometre of vertical uncertainty against a 1–2 km boundary layer constrains plume altitude but does not resolve surface coupling. Validated against a single spaceborne lidar with limited collocation statistics and no ground profile network. **Why it matters here:** vertical distribution is the dominant error term in AOD-to-surface-PM_{2.5} conversion, and smoke and dust are exactly where it bites.

doi: 10.5194/amt-19-5843-2026

Hu et al. 2026 Atmos Chem Phys

Biomass burning aerosols during the 2023 Canadian wildfires from dual-site Raman and fluorescence lidar

Dual-site observation of transported smoke, May–September 2023, at ATOLL (France, multi-wavelength Raman with a 466 nm fluorescence channel) and GPI (Russia, five-channel broadband fluorescence at 355 nm excitation). Upper-troposphere/lower-stratosphere layers showed higher depolarisation, slightly lower lidar ratios, lower Ångström exponents and a redshifted fluorescence peak against free-troposphere layers; fluorescence magnitude and spectral shape agreed across two sites and two instrument designs, the strongest evidence yet that these signatures are transferable rather than instrument-specific. The consequential finding is negative: **near-absence of hygroscopic growth at 90–100% RH**, contradicting the standard assumption that aged biomass-burning aerosol is hygroscopic — with direct consequences for every κ -Köhler humidity correction applied to optical PM sensors during smoke episodes. One fire season, two sites; altitude correlations at r^2 0.61–0.68 that the authors concede proxy for injection height and plume origin.

doi: 10.5194/acp-26-13027-2026

Ni et al. 2026 Atmos Meas Tech

An AI-based algorithm for retrieving aerosol optical depth and single scattering albedo from All-Sky Imager observations

Machine-learning retrieval trained on co-located sun-photometer AOD and SSA at two sites, Beijing_PKU in East China and SGP in the central US. Independent validation gave $r > 0.86$ for AOD at all wavelengths with RMSE below 0.25, and for SSA $r = 0.74$ (Beijing_PKU) and 0.84 (SGP) with RMSE below 0.09. Retrieving SSA from a consumer-grade sky camera is the striking part — absorption is normally the hardest aerosol property to retrieve and demands a calibrated multi-channel photometer. Two sites means two aerosol regimes and two instrument configurations, and an AOD RMSE of 0.25 is large wherever ambient AOD is low. As a densification strategy this is the same economics that produced PurpleAir — cheap hardware plus a trained model substituting for a reference instrument — with the same unanswered transferability question.

doi: 10.5194/amt-19-5871-2026

Zhao et al. 2026 PLoS One

Deep learning PM_{2.5} prediction integrating improved MSTL decomposition and the FATA optimizer

FATA-GSPMSTL-CNN-LSTM: RFECV plus Pearson feature screening, a grid-searched power-spectral variant of Multiple Seasonal-Trend decomposition using Loess, the Fata Morgana metaheuristic for hyperparameter search, and a CNN-LSTM head. Evaluated separately for heating and non-heating seasons in Guangzhou and Xianyang, and the seasonal split is the sensible part given how differently those two cities are sourced. Beyond that it follows a familiar pattern: four stacked components, **no ablation isolating which one carries the gain**, and improvement measured only against author-selected baselines. No prediction intervals, so the output is a point forecast with no stated uncertainty and cannot drive threshold alerting. Two cities, no external-city test.

doi: 10.1371/journal.pone.0358271

Peng et al. 2026 Environ Pollut

Inhibition of condensable particulate matter by co-firing biomass and high-sulfur coal

50 kW one-dimensional self-sustaining furnace, high-sulfur coal co-fired with corn stover or pine wood at 10% and 20% blend. Baseline condensable PM was 6.72 mg/MJ, 83.3% inorganic, falling to 5.09 and 3.85 mg/MJ for corn stover at 10% and 20% and to 5.36 and 4.55 for pine wood — **a 43% reduction at 20% corn stover**, traced to suppression of sulfate-associated inorganics and organic precursors. PAHs and polycyclic aromatic hydrocarbons fell below detection at 20% blending. **Why it matters here:** condensable PM is invisible to hot-filter methods and to every optical low-cost sensor, yet can rival filterable PM in mass, so inventories built on filterable-only sampling are biased low. One bench furnace, one coal, two feedstocks, no atmospheric ageing — and condensable PM measurement is method-dependent enough that absolute values should not travel across sampling protocols.

doi: 10.1016/j.envpol.2026.129163

Exposure assessment & modelling

4 records

Fireman et al. 2026 Int J Public Health

Biomonitoring of urban ultrafine particle exposure and associated health effects

99 hospital staff — 50 in the industrial Haifa Bay area, 49 Tel Aviv controls — with spirometry, FeNO, exhaled breath condensate and saliva at baseline and one year. Latent class analysis found a cluster with higher BMI, raised EBC ultrafine particle concentration and elevated salivary LDH-calcium, predominantly Haifa residents, reporting more cough, sputum and atopic complaints. The concept is worth tracking — EBC particle burden as an internal-dose biomarker would close the gap between a fixed monitor and what a person actually inhaled — but the execution does not yet support it. $n = 99$ with **LCA on a handful of variables will find clusters whether or not they exist**; no ambient UFP measurements accompany the biomarker, so the external-to-internal dose relation is never tested; and BMI loading on the same cluster makes an exposure reading hard to separate from a metabolic one.

doi: 10.3389/ijph.2026.1609855

Li et al. 2026 Atmosphere*A theoretical model for predicting visibility during dust events*

Physically based chain from surface wind erosion through dust emission and near-surface concentration to horizontal visibility, with an analytical concentration–visibility relation closed heuristically and a moment-balance entrainment model carrying median particle size and soil cohesion into the threshold friction velocity. Against three published datasets the relation explained $R^2 = 0.62$ and had the highest R^2 of the formulations compared under identical calibration. The appeal is economic: visibility is observed almost everywhere dust matters and PM₁₀ usually is not, so a defensible physical inversion turns an existing network into a concentration proxy. $R^2 = 0.62$ supports event detection and screening, not exposure assignment. The authors are explicit that **only the concentration–visibility link was evaluated** — the emission and entrainment stages are untested, and closing the chain needs synchronised observations that do not currently exist.

doi: 10.3390/atmos17090902

Paygude et al. 2026 Environ Monit Assess*Heavy metal exposure and source profiling from leaf-deposited particulate matter in Pune*

27 roadside, industrial, peri-urban, highway and rural sites. Elemental order Ca > K > Fe > Na > Mg > Mn > Zn > Cu > Cr > Pb, consistent with crustal material, construction and resuspended road dust; enrichment factors flagged Cu, Zn and Cr at traffic- and industry-influenced sites; PCA separated construction/road-dust, vehicular, biomass-or-food and localised industrial factors. Hazard quotients ordered Cr > Pb > Cu > Zn with hazard indices below 1 for adults and children. The method is the contribution: leaf-deposited dust is a zero-power, zero-maintenance passive sampler that resolves metal-enriched microenvironments a sparse reference network cannot. Its limits are equally clear — deposition velocity varies with leaf morphology, species and washoff history so concentrations are not convertible to airborne mass, integration time is uncontrolled, and an **inhalation hazard quotient computed from deposited dust requires an airborne concentration that is never observed here**.

doi: 10.1007/s10661-026-15906-w

Sen et al. 2026 Environ Microbiol Rep*Genomic and phenotypic classification of Staphylococcus species isolated from Delhi air*

Outdoor culturable bioaerosol sampling at six Delhi sites in November 2019 — the peak pollution month, pre-pandemic — using a Merck MAS Eco-100. *S. gallinarum*, *S. haemolyticus* and *S. arlettae* were recovered and characterised by whole-genome sequencing, phenotypic microarray, antibiotic-resistance profiling and biofilm formation in the presence of black carbon. Comparative genomics found the urease cluster *ureABCDEFG* in *S. arlettae* Naraina and *S. haemolyticus* Yojna Vihar 1, unusual because urease is normally reported negative in both species and may confer a niche fitness advantage. The black-carbon biofilm assay is the link to PM and the part worth following, but **no quantitative result for it is reported**. Culture-based sampling recovers a small biased fraction of the airborne microbiome, six sites in one month give no seasonal or spatial contrast, and no particle mass was measured alongside the biology, so the association with the pollution season is contextual.

doi: 10.1111/1758-2229.70411

Burden, policy & mitigation**3 records****Ju et al. 2026** Lancet Planet Health*Premature mortality and economic losses attributable to landscape fire-related PM_{2.5} in ten countries and territories*

1,366 communities, ten countries and territories, **47,844,851 deaths**, 2000–19, with community-specific associations estimated by instrumental-variable models and pooled by random-effects meta-analysis. Each 1 µg/m³ of 2-day mean landscape-fire PM_{2.5} raised mortality **0.60%** (95% CI 0.19–1.00) and mortality-related economic losses **0.59%** (0.21–0.97), yielding **906,869 deaths and US\$994.5 billion** attributable (attributable fractions 1.90% and 1.64%). The disproportion is the headline: fire PM_{2.5} is **19.0% of ambient mass but 30.6% of PM_{2.5}-attributable deaths**. IV estimates exceeded conventional association estimates by a **median 56.0%** (2.2–111.3), so classical designs are biased toward the null here. The weak point is that the instrument is not named in the abstract and exclusion-restriction validity — against co-emitted gases and weather — carries the causal claim entirely; value-of-statistical-life transfer across ten economies adds uncertainty that is not propagated into the intervals.

doi: 10.1016/j.lanplh.2026.101506

Bao et al. 2026 J Environ Manage*Distribution-sensitive assessment of air-quality and health benefits from clean-heating pilots in northern China*

88 clean-heating pilot cities, 2013–23, evaluated with machine-learning weather normalisation and the augmented synthetic control method against city-specific no-policy counterfactuals. The distributional result is the point: the policy cut the heating-season **mean only 3.01 µg/m³** but **P95 by 5.76, P99 by 5.96 and exceedance days by 7.92** per season — the tail moved roughly twice as far as the mean, which a mean-only evaluation misses entirely. Health impact assessment gave **12,454 fewer attributable all-cause deaths** (95% CI 8,564–16,222), with 88.9% from the 2017–18 rollout cohorts and 56.2% from the top decile of cities; high-pollution days were 27.8% of improvement days but **48.8% of the mortality difference**. Benefit is therefore concentrated in both time and space. ASCM counterfactuals depend on donor-pool comparability during a decade of nationwide decline, and the mortality figure is a transfer through an exposure-response function, not an observed death count.

doi: 10.1016/j.jenvman.2026.130960

Maighani et al. 2026 Rev Environ Health*An umbrella review of public health interventions to mitigate air pollution in high-income countries*

Eight systematic reviews spanning city and transport policy, household exposure reduction and emission-reduction modelling. Transport interventions gave consistent cardiovascular benefit and fewer traffic injuries; portable air cleaners cut indoor PM_{2.5} with modest blood-pressure improvement; modelling projected a **median ~1.5% mortality reduction** under large-scale mitigation. The ordering is the useful output — structural policy beat individual and household measures, which were complementary rather than substitutive. Two limits constrain its use. **Equity was rarely assessed and inconsistent where it was**, so the review cannot say who benefits; and the 1.5% figure is modelled rather than measured, mixing evidence tiers inside one summary. Eight reviews, all high-income, so none of it transfers to the settings carrying the largest burden.

doi: 10.1515/reveh-2026-0061

Respiratory & allergic**2 records**

Meng et al. 2026 Front Public Health

Exposure-response thresholds for ambient PM_{2.5} and SO₂ vs COPD and small airway dysfunction, Yunnan

11,095 residents aged 20+, multistage random cluster sampling 2021–22, with spirometry in questionnaire-screened high-risk individuals. COPD prevalence 27.2%, small airway dysfunction 53.6%. Restricted cubic splines gave significant nonlinearity for both pollutants with thresholds at 21.33 µg/m³ PM_{2.5} and 8.67 µg/m³ SO₂; above the PM_{2.5} threshold, COPD and SAD prevalence were 33.4% and 58.4%. Random-forest importance ranked air pollution above smoking and biomass fuel — a strong claim from a cross-sectional design that should be read as variable importance, not effect size. No temporality, exposure assigned without residential history, and the enriched high-risk sampling frame means these prevalences do not transfer to a general population. A 21 µg/m³ threshold sits well above the WHO 5 µg/m³ guideline and close to the regional mean, so it is plausibly a data-support artefact rather than a biological inflection.

doi: 10.3389/fpubh.2026.1874156

Vance et al. 2026 Scand J Public Health

Greenspace, air pollution and respiratory health outcomes: a systematic review of cohort studies

14 cohort studies to May 2026, all rated good on the Newcastle-Ottawa Scale, predominantly European. Greenspace was generally protective against air-pollution-associated respiratory mortality, and the methodological finding is the transferable one: **effects appeared more consistently through mediation than moderation**, implying greenspace acts by improving local air quality rather than by raising individual resilience. That has a direct modelling consequence — if the pathway is mediation, greenspace and PM_{2.5} should not be entered as independent covariates without care. No pooled estimate is reported, so the synthesis is narrative and magnitude is unavailable; greenspace and pollution are both strongly confounded by socioeconomic position; and near-complete European coverage limits transfer.

doi: 10.1177/14034948261481328

Occupational & indoor

2 records

Tam et al. 2026 Curr Environ Health Rep

Toxic air contaminant emissions from residential and commercial fossil gas appliances: a systematic review

25 studies, the first systematic synthesis of emission factors across gas appliance types. Nonzero emissions documented for **at least 41 toxic air contaminants** via both leakage and combustion pathways. Only formaldehyde and benzene had enough data for quantitative characterisation, and their emission factors **spanned two orders of magnitude** across studies and operating conditions — a spread that makes any single-value inventory estimate indefensible. Coverage is badly skewed toward cooking stoves; heaters, water heaters and dryers are barely studied despite longer duty cycles. The operationally interesting claim, correctly stated as emerging rather than established, is that **methane and CO measurements may proxy for leakage- and combustion-related TAC emissions** — which is exactly the kind of cheap-channel inference target a low-cost indoor node can carry.

doi: 10.1007/s40572-026-00562-6

Yoshizaki et al. 2026 J Am Coll Surg

Differences in the effectiveness of smoke evacuator types for reducing surgical smoke: a randomized controlled trial

Randomised, double-blind trial (UMIN000047033), 64 laparotomy patients at the University of Tokyo Hospital, Feb 2022 to May 2023, allocated 1:1:1 to electrocautery-incorporated evacuation at 8 mm tip distance ($n = 22$), 18 mm ($n = 23$) or hose-type ($n = 21$). The 8 mm system cut particle counts 96–97% across 0.3–5.0 µm against hose-type; **the 18 mm arm did not reach significance on particles**, so effectiveness collapses over a 10 mm change in capture distance — the practically important finding. Median acetaldehyde was 1.90, 2.70 and 7.90 µg/m³ in the 8 mm, 18 mm and hose arms, both electrocautery arms below hose, with formaldehyde behaving similarly. Endpoints are area concentrations rather than personal breathing-zone exposure and no health outcome was measured; single centre, one procedure type. The distance sensitivity is the transferable lesson for any point-source capture or near-field sensing geometry.

doi: 10.1097/xcs.0000000000001943

Mechanistic toxicology

2 records

Ma et al. 2026 Ecotoxicol Environ Saf

Combined PM_{2.5} and microplastics exposure exacerbates allergic asthma via OPA1/SLC7A11-mediated ferroptosis

Ovalbumin-induced murine asthma plus OVA-stimulated MRE cells exposed to PM_{2.5} and/or microplastics. Co-exposure amplified inflammatory infiltration, goblet-cell proportion and lung-function impairment beyond either agent alone, with raised ROS and MDA and reduced T-AOC, CAT and SOD. Ferroptosis markers moved coherently — Fe²⁺ accumulation, lipid peroxidation, NCOA4/FTH1/ACSL4 up, SLC7A11/GPX4/GCLC/GSS down — alongside mitochondrial depolarisation, ATP depletion and reduced respiratory complex activity. **OPA1 knockdown and ferrostatin-1 both attenuated the injury**, two independent interventions converging on one pathway, which is the strongest part of the design. What it cannot support: no dose-response against realistic ambient concentrations, and instillation or high-dose culture routinely exceeds inhaled doses by orders of magnitude. PM_{2.5} composition is unspecified, so this is one sample, not a class.

doi: 10.1016/j.ecoenv.2026.120808

Gulyaev et al. 2026 Biomed Rep

Polyphenols and inhalation-induced oxidative stress in chronic lung diseases: a systematic review

Systematic review to January 2026 across PubMed, Cochrane and Web of Science, with RoB 2 and SYRCLE risk-of-bias assessment. Preclinical evidence was consistent — attenuated oxidative damage, suppressed inflammatory signalling and enhanced antioxidant defence, largely via Nrf2 activation and NF-κB modulation, with histopathological improvement across several exposure models. **Clinical evidence was not**: some trials improved oxidative and inflammatory markers or symptom scores, but spirometric effects were inconsistent. That preclinical-to-clinical gap is the familiar signature of antioxidant intervention research, and the authors say so. Heterogeneity precluded meta-analysis, so there is no pooled estimate; trials were small and short. Recorded as the mitigation-side counterpart to the exposure literature, not as evidence that dietary intervention substitutes for reducing exposure.

doi: 10.3892/br.2026.2194

Neuro / mental health

1 record

Zhang et al. 2026 Front Public Health*Household solid fuel use, kitchen ventilation and depressive symptoms among older adults in China*

11,697 CLHLS participants aged 65+, 2018 wave. Depressive symptoms (CES-D-10) in **26.87%**, solid fuel use in 29.67%. Solid fuel carried **OR 1.25 (95% CI 1.09–1.44)** against clean fuel and **B = 0.57 (0.33–0.82)** on the continuous score — consistent but small, roughly a fifth of a scale point per unit. The charcoal subgroup showed a larger estimate on $n = 39$ exposed and the authors correctly flag it as unreliable. **No interaction was found between fuel type and kitchen ventilation**, which undercuts the paper's own framing that ventilation modifies the exposure. Exposure is entirely self-reported category with no PM_{2.5} measurement, so dose is unknown and misclassification is certain to be substantial; and the cross-sectional design cannot exclude reverse causation, since depressed older adults may differ in both household energy choice and reporting.

doi: 10.3389/fpubh.2026.1920233

Cardiovascular & metabolic

1 record

Shahriary et al. 2026 Toxicol Lett*Cardiovascular and pregnancy-related health effects of ultrafine particulate matter*

Mechanistic synthesis of experimental, controlled-human-exposure and epidemiological evidence linking ultrafine PM to cardiovascular injury, with emphasis on hypertensive disorders of pregnancy. Pathways covered: pulmonary and systemic inflammation, oxidative stress, mitochondrial dysfunction, autonomic imbalance, endothelial injury and disturbed angiogenic balance, extended to maternal cardiovascular adaptation and placental vascular development. The review's honest core is its attribution problem — it states plainly that **UFPM-specific findings are hard to separate from PM_{2.5}, diesel exhaust particles, engineered nanoparticles and traffic-related pollution generally**, and that direct developmental evidence is largely animal. Uncertainties named: particle translocation, exposure metric, critical gestational windows. No pooled estimates. The exposure-metric point is the operational one — UFP is a number-concentration phenomenon and mass-based networks, including nearly all low-cost optical nodes, do not measure it.

doi: 10.1016/j.toxlet.2026.113203

Reproductive & developmental

1 record

Ammons et al. 2026 Environ Res*Environmental chemical exposures and ovarian and endometrial cancers: a systematic review*

145 studies screened — 54% ovarian-cancer only, 53% North American, 40% cohort — with outdoor air pollution the single most studied exposure class (27 studies, 18%). The pooled PM_{2.5} result is a clean null for ovarian cancer: **HR 1.02 (95% CI 0.91–1.13)** per 5 µg/m³ across four studies, and **0.94 (0.81–1.09)** restricted to the three with co-pollutant adjustment. Endometrial cancer was weakly positive and implausibly precise — 1.02 (0.99–1.05) overall but **1.01 (1.00–1.01)** with co-pollutant adjustment and 1.01 (1.01–1.02) for incident cases only. **A confidence interval that narrow from five studies signals near-zero between-study heterogeneity or shared cohorts rather than genuine precision**, and no I^2 is reported. Worth recording as a negative: gynaecologic cancer is one endpoint where the PM_{2.5} signal does not appear, against a literature that increasingly reports positive associations wherever it looks.

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Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Zhang et al.	Associations between household solid fuel use, kitchen ventilation, and depressive symptoms among older adults in China: a national cross-sectional study.	<i>Front Public Health</i>	Cross-sectional (multistage)	Self-reported household fuel type and kitchen ventilation, no measured concentration	China
Cardiovascular & metabolic					
Shahriary et al.	The Silent Intruders: Cardiovascular and Pregnancy-Related Health Effects of Ultrafine Particulate Matter.	<i>Toxicol Lett</i>	Mechanistic review	Ultrafine particulate matter, number-based rather than mass-based metrics	Global / multi-region
Reproductive & developmental					
Ammons et al.	Epidemiologic research on environmental chemical exposures and ovarian and endometrial cancers: a systematic review.	<i>Environ Res</i>	Systematic review + random-effects meta-analysis	Long-term ambient PM _{2.5} , per 5 µg/m ³ , alongside other environmental chemical exposures	North America
Respiratory & allergic					
Meng et al.	Exposure-response relationship between ambient PM _{2.5} and SO ₂ with COPD and small airway dysfunction prevalence among individuals at high risk for COPD in Yunnan Province, China: a threshold analysis.	<i>Front Public Health</i>	Cross-sectional (multistage)	Ambient annual PM _{2.5} and SO ₂ , residential assignment	China
Vance et al.	Greenspace, air pollution and respiratory health outcomes: a systematic review of cohort studies.	<i>Scand J Public Health</i>	Systematic review	Residential greenspace (NDVI and similar) with co-assigned ambient air pollution	Europe
Mechanistic toxicology					
Gulyaev et al.	Association between polyphenols and inhalation-induced oxidative stress in chronic lung diseases: A systematic review.	<i>Biomed Rep</i>	Systematic review	Inhalation-induced oxidative stress across in vitro, animal and clinical exposure models	Global / multi-region
Ma et al.	Combined PM _{2.5} and microplastics exposure exacerbates allergic asthma via OPA1/SLC7A11-mediated airway epithelial ferroptosis.	<i>Ecotoxicol Environ Saf</i>	Animal + in vitro	PM _{2.5} and microplastic co-exposure, ovalbumin-sensitised murine model and MRE cells	China
Occupational & indoor					
Tam et al.	Toxic Air Contaminant Emissions from Residential and Commercial Fossil Gas Appliances: A Systematic Review.	<i>Curr Environ Health Rep</i>	Systematic review (PRISMA)	Appliance emission factors for 41 toxic air contaminants, leakage and combustion pathways	Global / multi-region
Yoshizaki et al.	Differences in the Effectiveness of Smoke Evacuators Types for Reducing Surgical Smoke: A Randomized Controlled Trial.	<i>J Am Coll Surg</i>	Randomised controlled trial	Optical particle counts at 0.3, 0.5, 1.0 and 5.0 µm plus acetaldehyde and formaldehyde, 10-min sampling	Japan
Sensing, forecasting & instrumentation					
Hu et al.	Advanced insights into biomass burning aerosols during the 2023 Canadian wildfires from dual-site Raman and fluorescence lidar observations.	<i>Atmos Chem Phys</i>	Remote sensing / lidar retrieval	Multi-wavelength Raman, depolarisation and fluorescence lidar profiles of transported smoke	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Li et al.	A multi-angle and polarization-based retrieval algorithm for aerosol layer height of smoke and dust.	<i>Atmos Meas Tech</i>	Satellite retrieval evaluation	Aerosol layer height from HARP2/PACE multi-angle polarimetry, validated against EarthCARE ATLID lidar	Global / multi-region
Ni et al.	An AI-based algorithm for retrieving aerosol optical depth and single scattering albedo using All-Sky Imager observations.	<i>Atmos Meas Tech</i>	Algorithm development	AOD and single scattering albedo retrieved from all-sky camera imagery, sun-photometer targets	Global / multi-region
Peng et al.	Inhibition of condensable particulate matter by co-firing biomass and high sulfur coal on a 50 kW self-sustaining combustion furnace.	<i>Environ Pollut</i>	Chamber emission characterisation	Condensable particulate matter mass and speciation, mg/MJ, dilution sampling	China
Zhao et al.	A novel deep learning prediction method for PM _{2.5} integrating improved MSTL decomposition and FATA optimization algorithm.	<i>PLoS One</i>	Deep-learning forecasting	PM _{2.5} time series with meteorological and pollutant co-variables, station-based	China
Exposure assessment & modelling					
Fireman et al.	Biomonitoring of urban ultrafine particle exposure and associated health effects.	<i>Int J Public Health</i>	Biomonitoring + in vitro	Ultrafine particle counts in exhaled breath condensate, industrial vs control urban setting	Israel
Li et al.	A Theoretical Model for Predicting Visibility During Dust Events.	<i>Atmosphere</i>	Theoretical / physical model	Horizontal visibility derived from saltation-controlled dust emission and near-surface concentration	Global / multi-region
Paygude et al.	Heavy metal exposure and source profiling based on leaf-deposited particulate matter in Pune city.	<i>Environ Monit Assess</i>	Biomonitoring + source apportionment	Leaf-deposited dust, ten elements, enrichment factors and PCA source profiles	India
Sen et al.	Genomic and Phenotypic Classification of Staphylococcus Species Isolated From Delhi Air.	<i>Environ Microbiol Rep</i>	Bioaerosol monitoring campaign	Culturable bioaerosol sampling (Merck MAS Eco-100), six sites, peak pollution season	India
Burden, policy & mitigation					
Bao et al.	Distribution-sensitive assessment of air-quality and health benefits from clean-heating pilots in northern China.	<i>J Environ Manage</i>	Augmented synthetic control + weather normalisation	City-daily PM _{2.5} from 1-km gridded product, ML weather-normalised, heating-season mean/P95/P99	China
Ju et al.	Premature mortality and economic losses attributable to landscape fire-related PM _{2.5} in ten countries and territories: a causal analysis.	<i>Lancet Planet Health</i>	Instrumental-variable time-series + random-effects meta-analysis	Population-weighted daily landscape-fire PM _{2.5} vs non-fire PM _{2.5} , 2-day moving average, modelled	Global / multi-region
Mairghani et al.	An umbrella review of public health interventions to mitigate air pollution and improve health outcomes in high-income countries.	<i>Rev Environ Health</i>	Umbrella review / meta-analysis	Intervention-level exposure change (traffic policy, portable air cleaners, net-zero pathways)	Global / multi-region

Provenance and method

Entry window **2026-09-16** → **2026-09-16**, contiguous with the 15 September issue; no gap. Built on 17 September under the standing rule that the current day is not opened before 22:00 local. **Sources.** PubMed [Date - Entry] run as two separate queries — health axis and instrumentation axis — through both the local harvester and the connector: **16 unique candidates**, the harvester a strict superset of the connector this time (16 vs 14), reversing the 15 September finding that neither client alone was sufficient. Europe PMC CREATION_DATE 56. Crossref windowed by ISSN across the tracked non-PubMed journals **66 deposits**, which carried all four remote-sensing records. **arXiv returned HTTP 406 again**; three header variants were tested this run (HTTPS with an Atom Accept, a browser user-agent, and plain HTTP) and all three failed identically, so the cause is not header negotiation and the leg should now be routed through a connector rather than repaired in `harvest.py`. OpenAlex HTTP 429, as on every recent run. **Consensus returned nothing admissible.** Two sweeps, 20 records: **seven were already in the archive** on a title check, and **all five that survived title screening and were then date-checked failed** — 10.3390/s26010280 created 2026-01-02, 10.1021/acs.est.6c00095 2026-05-08, 10.1016/j.scitotenv.2026.181915 2026-05-27, 10.1002/met.70228 2026-08-22, and the India AOD fusion work a December 2025 egusphere preprint. Consensus has no entry-date filter, so every result needs an independent Crossref `created` check; on this run the admitted yield was zero for the first time. **Europe PMC AGRICOLA defect.** Two AGR records returned inside a 16 September CREATION_DATE window resolve to **pubYear 2021 and 2014** — the Fishlake UAS emission-sampling paper and the Hodas outdoor-to-indoor transport model. The AGR leg retro-indexes, so a publication-year check is now mandatory before any AGR record is summarised. Both were rejected. **Screening.** 138 raw records across the four legs, 130 unique after cross-source dedupe, of which 8 were caught by the DOI gate against `seen.json`. **21 shipped, 27 excluded, 3 held.** The three held are same-day Elsevier deposits — Moscow long-term black carbon, the APR hybrid-imputation multi-target random forest, and the *Building and Environment* indoor CO₂ sensor-siting study — whose DOIs resolve but for which no abstract exists in Crossref, Europe PMC or Semantic Scholar. All three are on-axis and are deliberately kept out of `seen.json` so a later issue can carry them; this is the second consecutive issue in which the same Elsevier no-abstract-on-deposit pattern has cost tier-A candidates. **Trial watch.** Two registry records updated in window. NCT07540000 (INTERLUNG, Johns Hopkins, Bhaktapur, Nepal, PHASE4, $n = 800$) was already stored; endpoints re-checked and unchanged — pre-bronchodilator FEV₁ z-score at 40 months powered for 0.25 SD, plus SGRQ, the strongest genuinely respiratory outcome set in the accumulated file. NCT07793123 (Robertet SA, oral antioxidant vs placebo, skin radiance, $n = 60$) is new and kept, but is air-pollution-adjacent only: **all eight endpoints are dermatological or biophysical and no PM variable is measured** — the third consecutive topical hit whose PM framing no measured variable carries. `trials.json` was also **reconciled**: the running `trials` tally of 28 flagged on 15 September has been replaced by the derived count of persisted NCT records (12), and the counter must now be derived, never incremented. **Labels.** Subtopic, design, particle-metric and geography are assigned from title, abstract, keywords and MeSH; single-label by dominant contribution, a judgement rather than a controlled vocabulary. Five new design labels and one new geography key were registered in `plots.py` in the same edit that wrote them onto records. Effect estimates are transcribed verbatim from abstracts; the two landscape-fire percent changes are converted to risk-ratio equivalents and flagged as such in the forest-plot caption, and nothing else is rescaled. Content derived from PubMed and Crossref metadata; abstracts remain the copyright of their respective publishers.